

Exam Date & Time: 02-Dec-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Advanced Distributed systems [NPT-3206]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Consider a smart city IoT network consisting of thousands of interconnected sensors deployed across different areas to monitor environmental conditions like air quality, temperature, and traffic. These sensors operate autonomously, and the network must ensure that faults or failures are quickly detected without relying on a central monitoring server. Apply a gossip-based approach to detect faults in a smart city IoT network. (5)
 - A) ensure that faults or failures are quickly detected without relying on a central monitoring server. Apply a gossip-based approach to detect faults in a smart city IoT network.
 - B) The Network Setup for an application is as follows:
 - Central Index Server: Maintains a database of all files shared by peers.
 - Users: Connect to the server, register their file lists, and send search requests.
 - Peers: Directly transfer files with each other after receiving peer information from the central server. (3)

Let Alice wants to download a rare music file, "Song.mp3", from the Napster network. Identify the Step-by-Step Process for efficient management with the Central Index.

 - C) A distributed file-sharing platform is developed using the Chord protocol to manage the storage and retrieval of files across a network of peer nodes. The system must handle node failures and dynamic joins while ensuring that each file is accessible efficiently. Each node is responsible for a range of file keys based on their ID space. Files are stored at nodes using consistent hashing to maintain a uniform load across all nodes. Choose the hashing mechanism that distributes files across nodes in a Chord ring and explain. (2)
- 2) Apply the vector clocks to correctly order the following events across 3 processes and justify:
 - A) E1: Process A sends a message.
E2: Process B receives the message from A.
E3: Process B sends a message.
E4: Process A receives the message from B. (5)
 - B) Compare and contrast the Bully algorithm and the Ring algorithm in terms of message complexity, fault tolerance, and suitability for dynamic systems. Which algorithm would you recommend for a network with frequent node failures? Justify your answer. (3)

- C) In a weighted graph with nodes connected over unreliable communication links, describe how the GHS (Gallager-Humblet-Spira) algorithm ensures correct MST construction. Model a discussion on message passing and the phases of the algorithm. (2)
- 3) Consider a scenario where a social media platform uses eventual consistency for user-generated content. Analyze the potential challenges that may arise from this choice, especially regarding user experience and data accuracy. How can these challenges be mitigated? (5)
- A)
- B) Consider a scenario where multiple clients submit conflicting proposals in a Paxos-based system. Discuss how Paxos resolves conflicts and maintains a single consistent state. Analyze the implications for application-level behavior in this context. (3)
- C) Consider a real-world scenario in a blockchain network where some nodes may attempt to submit fraudulent transactions. Apply the Byzantine General's Problem to explain how consensus mechanisms (like Proof of Work or Proof of Stake) help maintain the integrity of the blockchain. (2)
- 4) Consider the implications of blockchain immutability in the context of Bitcoin transactions. Analyze the benefits and drawbacks of this property for users, regulators, and law enforcement agencies. How does immutability impact the resolution of disputes? (5)
- A)
- B) Facebook Cassandra employs a tunable consistency model. Identify the trade-offs between consistency, availability, and partition tolerance when configuring consistency levels in Cassandra. (3)
- C) A university uses Condor to manage computational biology simulations across high-performance servers and idle desktop machines. Analyze how Condor manages job scheduling in a distributed environment. (2)
- 5) In a financial trading system, multiple processes collaborate to execute trades across different markets. If one process responsible for updating stock prices fails during a volatile market session. Analyze the concept of virtual synchrony and how it helps maintain consistency in distributed systems during failures. (5)
- A)
- B) In what scenarios does DryadLINQ outperform traditional MapReduce frameworks? Analyze the advantages DryadLINQ offers in terms of flexibility and expressiveness. (3)
- C) Alice tries to double-spend the same 1 Bitcoin by sending it to Charlie. Apply proof-of-work (PoW) to maintain consensus in a decentralized network. (2)

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